

A Artifact Appendix

A.1 Abstract

This artifact provides the implementation and automated evaluation workflow for DeepStack’s analytical modeling and design-space exploration (DSE) of distributed 3D DRAM-stacked LLM accelerators. It reruns the fixed configurations underlying paper Figs. 8–12, Figs. 13–21, and Table 4; checks them against bundled, checksummed GPU and simulator references; and regenerates the paper-facing plots. No GPU, model weights, or external simulator is required.

A.2 Artifact check-list (meta-information)

- **Algorithm:** analytical latency, throughput, energy, thermal, and network modeling with hierarchical DSE.
- **Program:** Python 3.11 package and staged `run.sh` workflow.
- **Compilation/Binary:** no local compilation; three licensed, prebuilt CPython 3.11 x86-64 extensions are included.
- **Model:** DeepSeek-V3, Qwen3-235B-A22B, Llama-70B, and kernel-level workload descriptions; no model weights.
- **Data set:** checksummed GPU/vLLM and ASTRA-Sim/NS-3 references, fixed-configuration manifests, routing traces, and a recorded 301,728-row decode-DSE population.
- **Run-time environment:** x86-64 Linux, GNU Bash, Conda, CPython 3.11, and glibc 2.29 or newer.
- **Hardware:** 32 CPU cores and 64 GiB RAM recommended; no GPU, FPGA, or accelerator.
- **Metrics:** weighted MAPE, scaled tokens/s, bandwidth, speedup, temperature, and area feasibility.
- **Output:** validated CSVs, run/verification manifests, and 16 figure groups in PNG and PDF.
- **Experiments:** 15 fail-fast stages plus independent verification.
- **Disk space:** approximately 5 GiB including the environment.
- **Preparation time:** approximately 5–10 minutes.
- **Experiment time:** 12.2 minutes on the 64-core/128-thread packaging host with 32 workers; 30–45 minutes estimated on 32 cores.
- **Publicly available?:** GitHub and archived DOI.
- **Licenses:** Apache-2.0 for source and data; bundled extensions use LicenseRef-DeepStack-AE-Binary-1.0.
- **Automation:** repository-local scripts; no external framework.
- **Archived DOI:** [10.5281/zenodo.21351389](https://doi.org/10.5281/zenodo.21351389).

A.3 Description

A.3.1 How to access

Clone github.com/tile-ai/DeepStack (branch `ae`) or download the DOI archive above. The root `README.md` is the reviewer guide.

A.3.2 Hardware dependencies

Use a 64-bit x86 Linux host with 32 cores, 64 GiB RAM, and 5 GiB free storage. Lower core counts work with fewer workers and longer runtime. GPU and simulator measurements are bundled reference inputs and are not rerun.

A.3.3 Software dependencies

The artifact requires GNU Bash, Conda, CPython 3.11, and glibc 2.29 or newer. `./setup.sh` installs pinned dependencies. Three licensed prebuilt extensions provide protected model components; no compiler, CUDA, external simulator, or proprietary EDA tool is needed.

A.3.4 Data sets and models

All required references, configurations, routing traces, and plotting data are included and covered by SHA256SUMS. The artifact contains workload descriptions and routing metadata, but no prompts, weights, or hidden states. The complete area model and die-area breakdown are not distributed. Reference feasibility exposes only a feasible-SM count from a finite compiled capacity table.

A.4 Installation

```
./setup.sh
conda run --no-capture-output -n deepstack-ae \
./run.sh doctor
```

Installation succeeds when the second command prints `doctor: PASS`.

A.5 Experiment workflow

```
conda run --no-capture-output -n deepstack-ae \
./run.sh quick --model-version both
conda run --no-capture-output -n deepstack-ae \
./run.sh reproduce --workers 32 \
--model-version both
./verify.sh results/reproduce
```

Checked-in renderings under `prebuilt/figures/` are inspection-only previews. The workflow creates an independent result bundle under `results/reproduce/`; neither the model nor verifier reads the prebuilt directory.

Regenerated paper-facing plots use `paper_legacy`; `current_corrected` is separate diagnostic output. Paper Fig. 9 additionally audits the frozen paper statistic separately from its legacy rerun. No reported statistic mixes the two paths.

A.6 Evaluation and expected results

A successful full run prints `workflow: PASS` and `Overall verification: PASS`, then produces 57 validated CSVs, 16 status markers, and 16 figure groups. Artifact stage IDs retain the accepted-revision numbering: paper Figs. 11, 12, and 13–21 map to `fig12`, `fig13`, and `fig15–fig23`; the first three validation stages are unchanged.

- **Validation (paper Figs. 8–12):** H100 weighted MAPE 8.366%; B200 frozen-paper MAPE 12.918976% and legacy-rerun MAPE 12.724500%; MI325X mean per-model weighted MAPE 13.095%; sparse-attention legacy weighted MAPE 2.612%; and verified NoC closure including the reported 108,000× endpoint ratio.
- **Co-design (paper Figs. 13–21 and Table 4):** the workflow regenerates the selected winners and sweeps, including the nine-layer bandwidth peak, 39.62% decrease by layer 12, and Table 4 speedups of 2.7906× and 9.4744×.

Exact checks and tolerances are in each stage’s `summary.csv` and `docs/result_matrix.md`. The full $\sim 2.5 \times 10^{14}$ -point search and the non-distributable hardware-emulation reference are outside the reproduced scope.

A.7 Experiment customization

Custom architecture, energy, NoC, and DSE interfaces and runnable synthetic examples are documented in `README.md` and `tests/`. They include caller-owned capacity providers or area estimators.

A.8 Notes**-workers 16.**

On failure, inspect the stage log named in `run_manifest.csv`. For memory pressure, rerun with